

Keecas Quarto Example

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How to use keecas in a Quarto document

This example demonstrates all the key features of **keecas** for symbolic and units-aware calculations in a Quarto document.

Note

If the rendering engine is **KaTeX** (i.e. Jupyter notebook), the label command `\label{}` will result in a **ParseError**. The latex code will work when rendering by `quarto render`, but since you may want to see the result before rendering the document, an option to disable the `\label{}` command is provided.

When rendering with `quarto` you can have two options:

- if `qmd` files, then everything is fine
- if `ipynb` files, then pass the `--execute` flag to `quarto render` to rerender the notebook

Basic Symbolic Math with Units

Define symbols and calculate basic engineering quantities:

$$F = 10.00 \text{ kilonewton} \quad (\text{applied force}) \quad (1)$$

$$A = 50.00 \text{ cm}^2 \quad (\text{cross-sectional area}) \quad (2)$$

$$\sigma = \frac{F}{A} = 2.00 \text{ MPa} \quad (\text{normal stress}) \quad (3)$$

Pipe Commands Demonstration

Using pipe operators for functional composition:

$$x3.00 \text{ m} \quad (4)$$

$$y4.00 \text{ m} \quad (5)$$

$$d = x^2 + y^2 = 25.0 \text{ m}^2 \quad (6)$$

Different LaTeX Environments

Cases Environment

$$\begin{cases} E_{steel} = 200.00 \text{ gigapascal} & \text{steel elastic modulus} \\ E_{concrete} = 30.00 \text{ gigapascal} & \text{concrete elastic modulus} \end{cases} \quad (7)$$

Equation Environment with Labels

Calculate beam deflection with cross-references:

$$\delta = \frac{5 q L^4}{384 E I} = 15.949 \text{ mm} \quad (8)$$

i Note

labels emitted by `show_eqn` are latex labels, therefore they need to be referenced with `\eqref{}` or `\ref{}` latex commands.

The deflection calculated in 8 shows acceptable values for serviceability.

?@eq-QUARTO_EXAMPLE-delta

Verification Function

Using the `check` function for design checks:

$$\sigma_{Sd} 20.00 \text{ MPa} \quad (9)$$

$$\tau_{Sd} 15.00 \text{ MPa} \quad (10)$$

$$\sigma_{Rd} 250.00 \text{ MPa} \quad (11)$$

$$\tau_{Rd} 75.00 \text{ MPa} \quad (12)$$

$$\frac{\sigma_{Sd}}{\sigma_{Rd}} = 0.080 \quad [\leq 1.0 \text{ VERIFIED}] \quad (13)$$

$$\frac{\tau_{Sd}}{\tau_{Rd}} = 0.200 \quad [\leq 1.0 \text{ VERIFIED}] \quad (14)$$

The type of comparison in the `check` function can be specified, as well the value to be specified against.

$$a = 3 \quad [> 1 \text{ NOT VERIFIED}] \quad (15)$$

$$b = 4 \quad [> 2 \text{ VERIFIED}] \quad (16)$$

$$c = 5 \quad [\geq 3 \text{ NOT VERIFIED}] \quad (17)$$

$$d = 6 \quad [\geq 4 \text{ VERIFIED}] \quad (18)$$

$$f = 7 \quad [\neq 5 \text{ NOT VERIFIED}] \quad (19)$$

Subs get ordered

Substitution will get ordered and sorted automatically, so you can use a symbol before mapping to a value.

$$a = 3 \tag{20}$$

$$b = 4 \tag{21}$$

$$d = \frac{\sqrt{a^2 + b^2}}{c} = 0.417 \tag{22}$$

$$c = a \, b = 12.000 \tag{23}$$

Summary

This example demonstrated:

- Basic symbolic math with units
- Pipe command usage (see Section)
- Different LaTeX environments (align, cases, equation)
- Label and cross-reference functionality
- Dataframe operations
- Verification functions
- Complex symbolic manipulations

All calculations in Section show that keecas provides a powerful interface for engineering calculations in Quarto documents.